

# CSMFAB000000000107 Aegis Concrete Formulation and Pumpability

## V1.0 to V2.0 Versioning Delta

**Type:** Scientific Change Log | **Date:** June 2026 | **Applies to document:** CSMFAB000000000107 Aegis Concrete Formulation and Pumpability V2.0

### Revision Necessity — Three Drivers

- 1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
- 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
- 3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

### V1.0 vs V2.0 — Specific Parameter Changes

| Parameter                   | V1.0                                               | V2.0                                                                                             | Reason                                                                         | Primary Source                                      |
|-----------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------|
| Solar Cycle 25 risk framing | V1.0: Risk basis SSN ~115 (original SC25 forecast) | V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20% | Observed solar activity exceeded forecast; design basis was non-conservative   | NOAA SWPC Solar Cycle 25 Progression Report Q1 2026 |
| Concrete binder             | V1.0: Portland cement OPC; ~505 kg CO2/m3; 40 MPa  | V2.0: Geopolymer binder; ~113 kg CO2/m3; 80 MPa (RILEM 2025)                                     | 78% CO2 reduction; 2x strength; ACI 318-25 geopolymer provisions now available | RILEM TC-281 2025; ACI 318-25; ASTM C1920-2024      |

| Parameter                             | V1.0                                        | V2.0                                                             | Reason                                                                                  | Primary Source                                    |
|---------------------------------------|---------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>BFRP rebar design standard</b>     | V1.0: ACI 440.1R-15                         | V2.0: ACI 440.11-25 (Seismic Type S); design tensile 1000 MPa    | Seismic provisions absent in V1.0 — required for CA/PNW deployment; 11-year code update | ACI 440.11-25 January 2025; ACI 318-25            |
| <b>ZrB2-SiC sintering temperature</b> | V1.0: SPS at 1900–2100°C, 30–50 MPa, 20 min | V2.0: Flash Sintering at 1800°C, 50 MPa, 5 min; >98% TD achieved | 40% energy reduction; lower T preserves FSS layer integrity; equivalent density         | Johnson et al., Acta Materialia 278 (2024) 120223 |

### Impact If V1.0 Retained (Criticality Matrix)

| Parameter Changed              | Consequence of Non-Upgrade                                                           | Severity |
|--------------------------------|--------------------------------------------------------------------------------------|----------|
| Solar Cycle 25 risk framing    | Probability underestimated; design basis not conservative for observed SC25 activity | MEDIUM   |
| Concrete binder                | 505 vs 113 kg CO2/m3; sustainability mandate unmet; lower strength ceiling           | MEDIUM   |
| BFRP rebar design standard     | Non-compliant with SDC D/E/F seismic zones; limits geographic deployment             | HIGH     |
| ZrB2-SiC sintering temperature | Excess 400°C degrades embedded FSS; higher energy cost vs V2.0 process               | MEDIUM   |

### Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

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*End of Versioning Delta | CSMFAB000000000107 Aegis Concrete Formulation and Pumpability V1.0 archived; V2.0 is the active specification as of June 2026.*